

Reflection

Since participation on Grad Café is entirely voluntary, the data has self-selection bias which means individuals with high score and strong credentials or top-tier acceptances are more willing to post their outcome. For example, in my analysis about question4, which revealed an average GPA of 3.66 among American applicants for Fall 2026. This results maybe too high because that applicants with lower GPA or who received unreturned applications or rejections are significantly less likely to publicly record their outcomes, results in overestimate of the average data.

Moreover, the dataset is plagued by pervasive e missing data, unstandardized text entries and noise which may because when people recorded their data, they are not following the standard way. Which create a sharp gap between what the database mathematically contains and what conclusions can be obtained about the real world. Take question 10 as an example, when I calculated group-level statistic, I need to filter out rows with missing values, which make my result excludes large amount of data from applicants who may has different GPA and results in bias.

Besides, the inconsistency in raw user entries, like they can use different abbreviations of their university and program, will cause bias.

Overall, while SQL query results provide mathematically calculations of data stored on disk using accurate code logical, it cannot represent the real-world situation